Punjab University Lahore (2022)
Math (201)
Question #01

i) All vector space must be abelian group under multiplication. Justify your answer with example.

In a vector space, the scalar multiplication is defined between a scalar and a vector, not between two vectors.

(e.g) $V = \mathbb{R}^2$

$a(x, y) = (ax, ay)$ where, a is a scalar. $\mathbb{R}^2$

$(x, y) + (z, w) = (x+z, y+w)$

$(x, y) + (z, w) = (z, w) + (x, y)$

Therefore a vector space is an abelian group under addition, but

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not under multiplication.

ii) Show that any linearly independent set of vector of any vector space can be extended to a basis for V.

Let V be a vector space and let $S = \{v_1, v_2, v_3, \ldots, v_n\}$ be a linearly independent set of vectors in V. We want to show that S can be extended to basis for V.

Step 1:- If S is already a spanning vector for V, then it is a basis for V.

Step 2:- If S is not a spanning set for V, then there exists a vector v in V that cannot be written as a linear combination of vector in S.

Step 3:- Let $S' = S \cup \{v\}$. We claim that S' is still linearly independent.

Suppose $(a_1, a_2, \ldots, a_n)$ and (b).

$a_1 v_1, a_2 v_2, \ldots, a_n v_n + bv = 0$

Therefore, any linear independly

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set of vector in a vector space can be extended to basis for the vector space.

iii) Let V be a vector space over any field and $W \subseteq V$. Then W is subspace of V or not?

let $V = \mathbb{R}^3$

$W = \{(x, y, z) \mid x^2 + y^2 + z^2 = 25\} \subseteq \mathbb{R}^3$

$(4, 3, 0) \in W \subseteq \mathbb{R}^3$

but $k = 2 \in \mathbb{R}$

$k(4, 3, 0) = 2(4, 3, 0)$

$= (8, 6, 0)$

$8^2 + 6^2 + 0^2 \neq 25$

$k(4, 3, 0) \notin W$

So, W is not subspace of $\mathbb{R}^3$.

iv) Define:-

Orthogonal matrix:- Orthogonal matrix preserve the Euclidean norm and are widely used in linear algebra, statics and many areas of science and

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engineering.

$Q^T = Q^{-1}$ (the transpose of Q is inverse)

$Q^T Q = Q Q^T = I$ (identity matrix).

Similar matrix:- Similar matrix are matrixs that have the same eigenvalues and can be transformed into each other through a series of row and coloumn operations.

$$B = P^{-1} A P$$

V) **Define:-**

Linear dependence:- Linear dependence refers to the relationship between a set of vectors where one or more vectors can be expressed as a linear combination of the others.

$$\{a_1 v_1 + a_2 v_2 + \cdots + a_n v_n\} = 0$$

Linear independence:- Linear independence is a concept in linear algebra that describes

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a set of vector that cannot be apressed as a linear combination of each vector.

$\{v_1, v_2 - - - - , v_n\}$

$\{a_1 v_1, a_2 v_2, \text{———}, a_n v_n\}$.

vi) **Define:-**

determinants:- Determinants are scalar value that can be computed from the elements of a square matrix.

Cofectors & minors:- Cofectors are the element of a matrix that are used to compute the determinant of the matrix. They are defined as the minor of the matrix multiplied by sign factor. $C_{ij} = (-1)^{(i+j)} * M_{ij}$

Question # 02

Long Questions:-

-1) let A and B be two matrix of order 6 such that $\det(AB)^2 = 72$

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$\det(A^2 B^2) = 144$ Find ~~[illegible]~~.
$\det(A)$, $\det(2A)$ and
$\det(AB^0)$

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2) Let $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ defined
$T(x_1, x_2, x_3) = (x_1, -x_2, x_3)$
Is T is linear. Find $N(T)$ and
$R(T)$

Ex:- 6.4 Question # 12

3) Find solution by Gauss-Jordan Elimination method.

$2x_1 - x_2 - x_3 = 4$, $3x_1 - 2x_2 + 4x_3 = 11$,
$3x_1 - 2x_2 + 4x_3 = 11$

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4) Find eigen values and corresponding Eigen vectors of matrix $A = \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$

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1) Find characterics eq. $\det(A - \lambda I) = 0$

2) Solve λ (eigen values)

3) Find eigen vectors $(A - \lambda I)x = 0$.

1) $\det(A - \lambda I) = \det \begin{bmatrix} 5-\lambda & 9 & 3 \\ 3 & 5-\lambda & 6 \\ -1 & -5 & -3-\lambda \end{bmatrix}$

$= (5-\lambda)(-(5-\lambda)(3-\lambda) - 30) - 9(-9 - 3(3+\lambda))$

$-3(-15 + 5(3+\lambda))$

$= (5-\lambda)(15 - 8\lambda + \lambda^2) + 81 + 27\lambda + 9\lambda^2$

$+45 - 75\lambda + 15\lambda^2$

$= 15 - 8\lambda + \lambda^2 - 75\lambda + 81 + 27\lambda - 15\lambda^2$

$+45 - 75\lambda + 15\lambda^2$

$= -30\lambda^2 - 121\lambda + 141$

$= 0$

Solve λ (eigenvalues):-

$-30\lambda^2 - 121\lambda + 141 = 0$ Divide by -30

$\lambda^2 + 4.033\lambda - 4.7 = 0$

Solve the quadratic eq:-

$\lambda = -6.033$, $\lambda = 0.767$.

Find the eigen vectors:-

$\lambda = -6.033$

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$(A - \lambda I)x = 0$

$$\begin{bmatrix} 5+6.033 & 9 & 3 \\ -3 & 5+6.033 & 6 \\ -1 & -5 & -3+6.033 \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 0.7071 \\ -0.7071 \\ 0 \end{bmatrix}$$

for $\lambda = 0.767$

$(A - \lambda I)x = 0$

$$\begin{bmatrix} 5-0.767 & 9 & 3 \\ 3 & 5-0.767 & 6 \\ -1 & -5 & -3-0.767 \end{bmatrix} x = 0$$

$$x = \begin{bmatrix} 0.5774 \\ 0.5774 \\ 0.5774 \end{bmatrix}$$

So, eigenvalues are $\lambda = -6.033$

and $\lambda = 0.767$.

eigen vectors: $\begin{bmatrix} 0.7071 \\ -0.7071 \\ 0 \end{bmatrix}$,

$$\begin{bmatrix} 0.5774 \\ 0.5774 \\ 0.5774 \end{bmatrix}$$

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5) What condition must a, b, c and d satisfy so, that the following matrix are linearly independent.

$M_{22} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

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